

LINEAR REGRESSION

```
In [121... import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

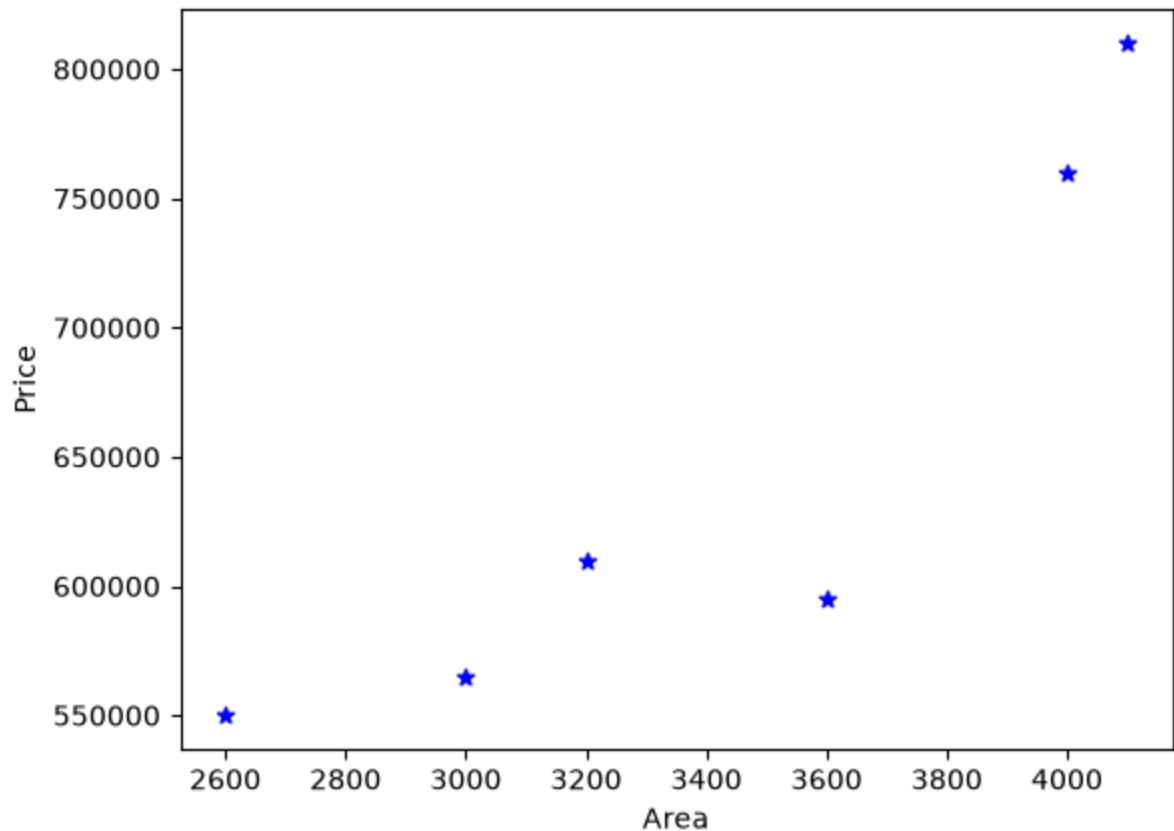
```
In [122... df = pd.read_csv("homeprices.csv")
df
```

```
Out[122...   area  bedrooms  age  price
0  2600         3.0   20  55000
1  3000         4.0   15  565000
2  3200         NaN   18  610000
3  3600         3.0   30  595000
4  4000         5.0    8  760000
5  4100         6.0    8  810000
```

To just see

```
In [123... %matplotlib inline
plt.xlabel("Area")
plt.ylabel("Price")
plt.scatter(df.area, df.price, color = "blue", marker = "*")
```

```
Out[123... <matplotlib.collections.PathCollection at 0x17b6f9aafd0>
```



Step 3: Prepare the Data (Features and Target)

Separate your dataset into features

The **independent variables** - **area**

The **target** - want to **predict**, e.g., **price**).

In [124...

#Note: Scikit-Learn expects the feature variable (X) to be a 2D array or DataFrame,

```
# Feature(s) - must be 2D (DataFrame)
X = df[['area']]
X
```

Out[124...

	area
0	2600
1	3000
2	3200
3	3600
4	4000
5	4100

```
In [125]: # Target variable - 1D (Series)
y = df['price']
y
```

```
Out[125]: 0    550000
          1    565000
          2    610000
          3    595000
          4    760000
          5    810000
          Name: price, dtype: int64
```

```
In [126]: # Create the model
model = LinearRegression()

# Train (fit) the model using your data
model.fit(X, y) #print("Model trained successfully!")
```

```
Out[126]: ▼ LinearRegression ⓘ ?
          ► Parameters
          ► Fitted attributes
```

```
In [72]: # Example: Predict the price for a home with an area of 3300 sq. ft.
new_area = pd.DataFrame([[3300]], columns=['area']) # Using a DataFrame keeps featu
new_area
```

```
Out[72]:    area
0    3300
```

```
In [73]: predicted_price = model.predict(new_area)

print(f"Predicted Price: ${predicted_price[0]:,.2f}")
```

Predicted Price: \$628,715.75

Generate CSV file with list of home price predictions

```
In [117]: area_df = pd.read_csv("areas.csv")
area_df
```

Out[117...

	area
0	1000
1	1500
2	2300
3	3540
4	4120
5	4560
6	5490
7	3460
8	4750
9	2300
10	9000
11	8600
12	7100

In [118...

```
model.predict(area_df).round(2)
```

Out[118...

```
array([ 316404.11,  384297.95,  492928.08,  661304.79,  740061.64,  
        799808.22,  926090.75,  650441.78,  825607.88,  492928.08,  
        1402705.48, 1348390.41, 1144708.9 ])
```

In [119...

```
area_df['prices'] = model.predict(area_df).round(0).astype(int)  
area_df
```

Out[119...

	area	prices
0	1000	316404
1	1500	384298
2	2300	492928
3	3540	661305
4	4120	740062
5	4560	799808
6	5490	926091
7	3460	650442
8	4750	825608
9	2300	492928
10	9000	1402705
11	8600	1348390
12	7100	1144709

In [120...

```
area_df.to_csv('prediction.csv', index= False)
```